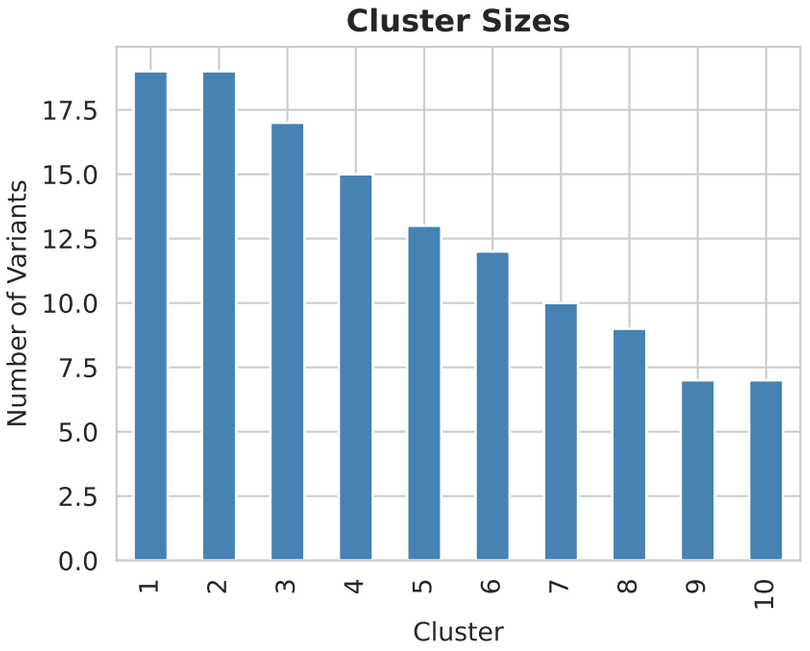
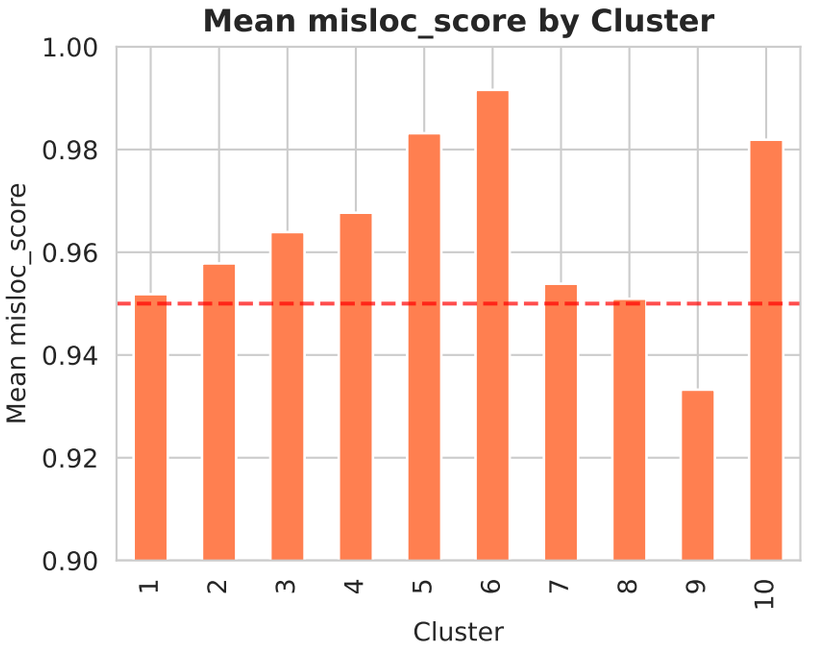
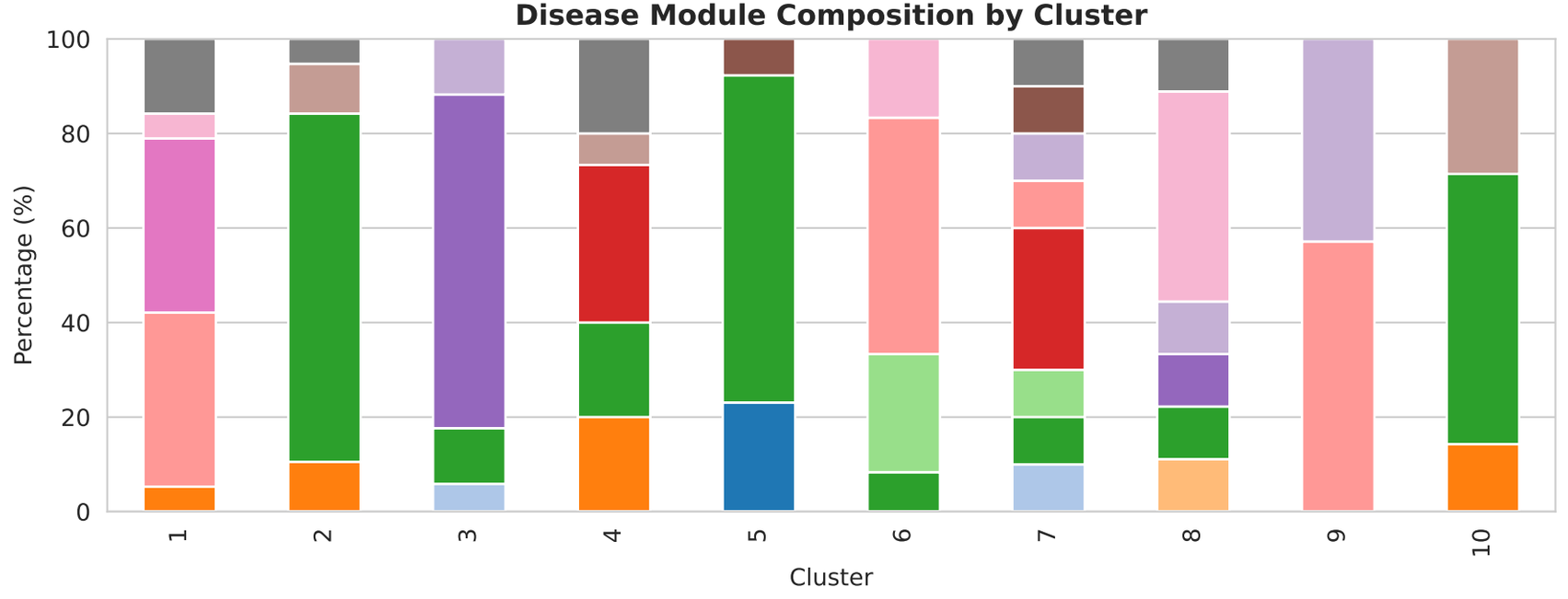
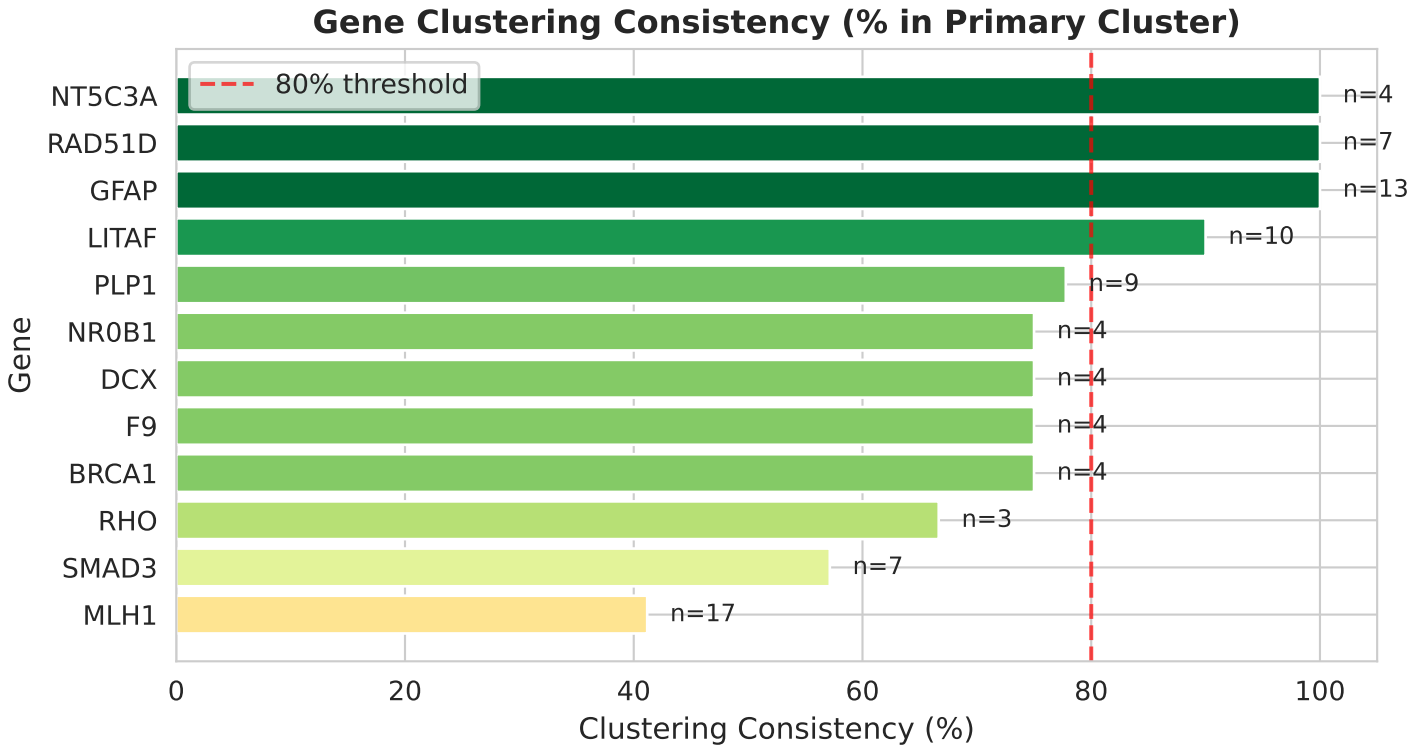
Comprehensive Disease Biology Analysis - Overview

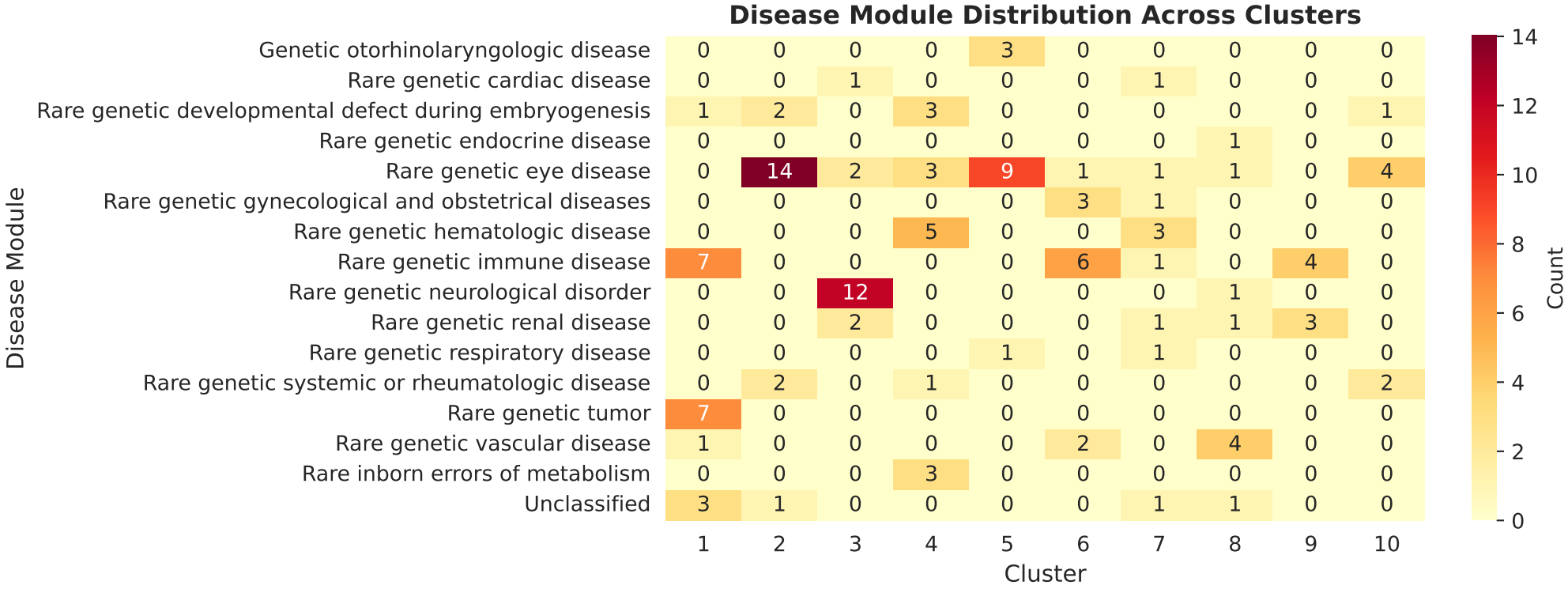






Gene and Disease Module Patterns





Cluster 1: Comprehensive Analysis

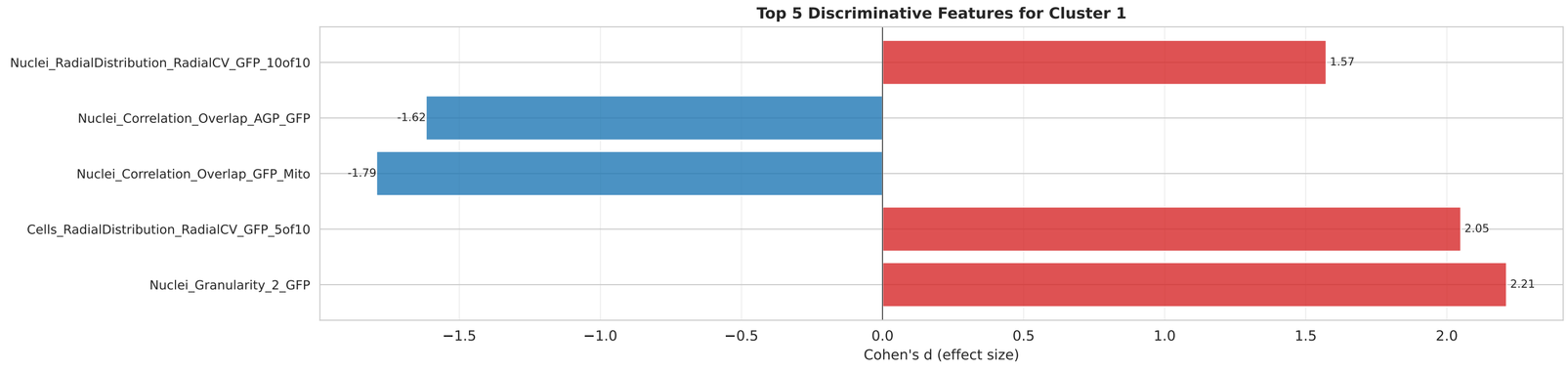
```
CLUSTER 1 ANALYSIS
================================================================================================

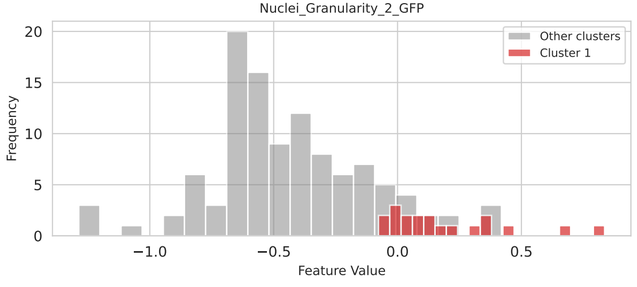
Size: 19 variants
Mean misloc_score: 0.952
Hit variants: 13 (68.4%)

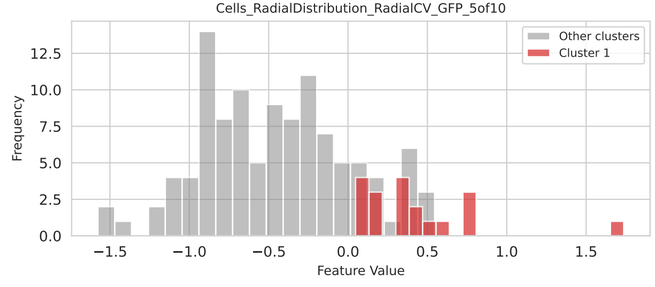
ALL VARIANTS (sorted by gene, position):
--------------------------------------------------
  KRT8_Tyr54His           |  KRT8_Ile63Val          |  KRT8_Gly434Ser         |  MLH1_His318Gln
  MLH1_Ser508Arg          |  MLH1_Lys618Arg         |  MLH1_Pro640Ser         |  MLH1_Lys670Arg
  MLH1_Cys680Arg          |  MLH1_Ala681Thr         |  RAD51D_Ile17Val        |  RAD51D_Val132Ile
  RAD51D_Asn138His        |  RAD51D_Ala231Ser       |  RAD51D_Phe305Leu       |  RAD51D_Met308Val
  RAD51D_Thr328Ile        |  SMAD3_Val410Phe        |  ZC4H2_Leu66His

DISEASE MODULES (detailed):
--------------------------------------------------
  Rare genetic tumor: 14 [MLH1_Ala681Thr, MLH1_Pro640Ser, MLH1_Cys680Arg, MLH1_Lys618Arg, RAD51D_Phe305Leu, RAD51D_Val132Ile, RAD51D_Asn138His, RAD51D_Met308Val, RAD51D_Ala231Ser, RAD51D_Thr328Ile, RAD51D_Ile17Val, MLH1_Lys670Arg, MLH1_Ser508Arg, MLH1_His318Gln]
  Rare genetic immune disease: 7 [MLH1_Ala681Thr, MLH1_Pro640Ser, MLH1_Cys680Arg, MLH1_Lys618Arg, MLH1_Lys670Arg, MLH1_Ser508Arg, MLH1_His318Gln]
  Inherited cancer-predisposing syndrome: 7 [MLH1_Ala681Thr, MLH1_Pro640Ser, MLH1_Cys680Arg, MLH1_Lys618Arg, MLH1_Lys670Arg, MLH1_Ser508Arg, MLH1_His318Gln]
  Rare genetic developmental defect during embryogenesis: 1 [ZC4H2_Leu66His]
  Rare genetic neurological disorder: 1 [ZC4H2_Leu66His]
  Rare genetic vascular disease: 1 [SMAD3_Val410Phe]
  Rare genetic systemic or rheumatologic disease: 1 [SMAD3_Val410Phe]

ANNOTATION ENRICHMENTS (values appearing ≥2 times):
--------------------------------------------------
  Disease: LYNCH SYNDROME 2 (7x) [MLH1_Ala681Thr, MLH1_Ala681Thr, MLH1_Pro640Ser, MLH1_Cys680Arg, MLH1_Lys618Arg, MLH1_Ser508Arg, MLH1_His318Gln]
  Disease: BREAST-OVARIAN CANCER, FAMILIAL, SUSCEPTIBILITY TO, 4 (7x) [RAD51D_Phe305Leu, RAD51D_Val132Ile, RAD51D_Asn138His, RAD51D_Met308Val, RAD51D_Ala231Ser, RAD51D_Thr328Ile, RAD51D_Ile17Val]
  Disease: MUIR-TORRE SYNDROME (2x) [MLH1_Ala681Thr, MLH1_Ala681Thr]
  Disease: HEPATITIS C VIRUS, SUSCEPTIBILITY TO (2x) [KRT8_Tyr54His, KRT8_Ile63Val]
  inheritance_pattern: autosomal dominant (12x) [MLH1_Pro640Ser, MLH1_Cys680Arg, MLH1_Lys618Arg, RAD51D_Phe305Leu, RAD51D_Val132Ile, RAD51D_Asn138His, RAD51D_Met308Val, RAD51D_Ala231Ser, RAD51D_Thr328Ile, RAD51D_Ile17Val, MLH1_Ser508Arg, MLH1_His318Gln]
  inheritance_pattern: unknown (5x) [KRT8_Tyr54His, KRT8_Gly434Ser, KRT8_Ile63Val, MLH1_Lys670Arg, SMAD3_Val410Phe]
  pLDDT_Category: >90 (14x) [MLH1_Ala681Thr, ZC4H2_Leu66His, MLH1_Pro640Ser, MLH1_Cys680Arg, MLH1_Lys618Arg, RAD51D_Phe305Leu, RAD51D_Val132Ile, RAD51D_Asn138His, RAD51D_Met308Val, RAD51D_Ala231Ser, RAD51D_Ile17Val, MLH1_Lys670Arg, MLH1_Ser508Arg, SMAD3_Val410Phe]
  pLDDT_Category: ⩽50 (4x) [KRT8_Tyr54His, KRT8_Gly434Ser, KRT8_Ile63Val, RAD51D_Thr328Ile]
  RSA_Category: 10-90 (9x) [ZC4H2_Leu66His, KRT8_Gly434Ser, MLH1_Lys618Arg, RAD51D_Phe305Leu, RAD51D_Asn138His, RAD51D_Met308Val, MLH1_Lys670Arg, MLH1_Ser508Arg, MLH1_His318Gln]
  RSA_Category: <10 (7x) [MLH1_Ala681Thr, MLH1_Pro640Ser, MLH1_Cys680Arg, RAD51D_Val132Ile, RAD51D_Ala231Ser, RAD51D_Ile17Val, SMAD3_Val410Phe]
  RSA_Category: ⩾90 (3x) [KRT8_Tyr54His, KRT8_Ile63Val, RAD51D_Thr328Ile]
```







Cluster 2: Comprehensive Analysis

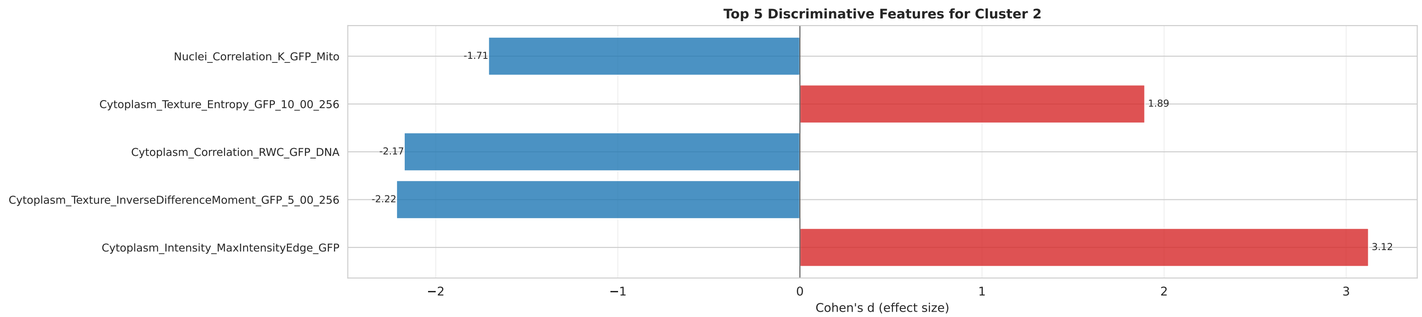
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CLUSTER 2 ANALYSIS
================================================================================

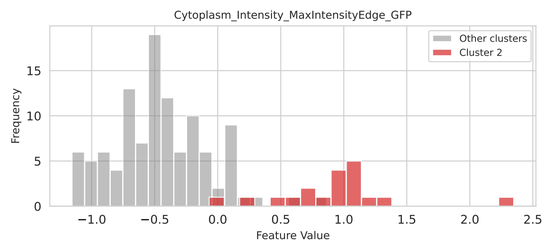
Size: 19 variants
Mean misloc_score: 0.958
Hit variants: 16 (84.2%)

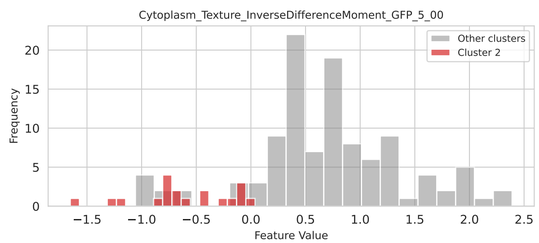
ALL VARIANTS (sorted by gene, position):
--------------------------------------------------
  APOA1_Trp74Arg          |  APOA1_Val180Glu         |  COMP_Asp319Val          |  GFAP_Asn77Ser
  GFAP_Arg79Cys           |  GFAP_Arg79Ser           |  GFAP_Arg88Cys           |  GFAP_Leu90Pro
  GFAP_Leu235Pro          |  GFAP_Lys236Thr          |  GFAP_Arg239Leu          |  GFAP_Arg239Pro
  GFAP_Lys279Glu          |  GFAP_Leu352Pro          |  GFAP_Tyr366Cys          |  GFAP_Glu371Gly
  SHOC2_Ser2Gly           |  SOS2_Val869Ile          |  ZC4H2_Arg18Lys

DISEASE MODULES (detailed):
--------------------------------------------------
  Rare genetic eye disease: 14 [GFAP_Asn77Ser, GFAP_Arg79Cys, GFAP_Leu352Pro, GFAP_Arg239Leu, GFAP_Arg88Cys, SHOC2_Ser2Gly, GFAP_Arg79Ser, GFAP_Tyr366Cys, GFAP_Arg239Pro, GFAP_Leu90Pro, GFAP_Leu235Pro, GFAP_Lys279Glu, GFAP_Lys236Thr, GFAP_Glu371Gly]
  Rare genetic neurological disorder: 14 [GFAP_Asn77Ser, GFAP_Arg79Cys, GFAP_Leu352Pro, GFAP_Arg239Leu, GFAP_Arg88Cys, ZC4H2_Arg18Lys, GFAP_Arg79Ser, GFAP_Tyr366Cys, GFAP_Arg239Pro, GFAP_Leu90Pro, GFAP_Leu235Pro, GFAP_Lys279Glu, GFAP_Lys236Thr, GFAP_Glu371Gly]
  Rare genetic developmental defect during embryogenesis: 3 [SHOC2_Ser2Gly, ZC4H2_Arg18Lys, COMP_Asp319Val]
  Rare genetic systemic or rheumatologic disease: 2 [APOA1_Trp74Arg, APOA1_Val180Glu]
  Rare genetic renal disease: 2 [APOA1_Trp74Arg, APOA1_Val180Glu]
  Rare genetic endocrine disease: 2 [APOA1_Trp74Arg, APOA1_Val180Glu]
  Rare inborn errors of metabolism: 2 [APOA1_Trp74Arg, APOA1_Val180Glu]
  Rare genetic cardiac disease: 1 [SHOC2_Ser2Gly]
  RASopathy: 1 [SHOC2_Ser2Gly]
  Rare genetic skin disease: 1 [SHOC2_Ser2Gly]

ANNOTATION ENRICHMENTS (values appearing ≥2 times):
--------------------------------------------------
  Disease: ALEXANDER DISEASE (12x) [GFAP_Asn77Ser, GFAP_Arg79Cys, GFAP_Leu352Pro, GFAP_Arg239Leu, GFAP_Arg88Cys, GFAP_Tyr366Cys, GFAP_Arg239Pro, GFAP_Leu90Pro, GFAP_Leu235Pro, GFAP_Lys279Glu, GFAP_Lys236Thr, GFAP_Glu371Gly]
  Disease: NOONAN SYNDROME-LIKE DISORDER WITH LOOSE ANAGEN HAIR 1 (2x) [SHOC2_Ser2Gly, SHOC2_Ser2Gly]
  inheritance_pattern: autosomal dominant (13x) [GFAP_Asn77Ser, GFAP_Arg79Cys, GFAP_Leu352Pro, GFAP_Arg239Leu, GFAP_Arg88Cys, SOS2_Val869Ile, GFAP_Tyr366Cys, GFAP_Arg239Pro, GFAP_Leu90Pro, GFAP_Leu235Pro, GFAP_Lys279Glu, GFAP_Lys236Thr, GFAP_Glu371Gly]
  inheritance_pattern: unknown (3x) [APOA1_Trp74Arg, GFAP_Arg79Ser, COMP_Asp319Val]
  pLDDT_Category: >90 (13x) [GFAP_Asn77Ser, GFAP_Arg79Cys, GFAP_Leu352Pro, GFAP_Arg239Leu, GFAP_Arg88Cys, ZC4H2_Arg18Lys, SOS2_Val869Ile, GFAP_Arg79Ser, COMP_Asp319Val, GFAP_Tyr366Cys, GFAP_Arg239Pro, GFAP_Leu90Pro, GFAP_Lys279Glu]
  pLDDT_Category: 70-90 (4x) [APOA1_Val180Glu, GFAP_Leu235Pro, GFAP_Lys236Thr, GFAP_Glu371Gly]
  RSA_Category: 10-90 (18x) [GFAP_Asn77Ser, APOA1_Trp74Arg, APOA1_Val180Glu, GFAP_Arg79Cys, GFAP_Leu352Pro, GFAP_Arg239Leu, GFAP_Arg88Cys, SHOC2_Ser2Gly, ZC4H2_Arg18Lys, GFAP_Arg79Ser, COMP_Asp319Val, GFAP_Tyr366Cys, GFAP_Arg239Pro, GFAP_Leu90Pro, GFAP_Leu235Pro, GFAP_Lys279Glu, GFAP_Lys236Thr, GFAP_Glu371Gly]
```







Cluster 3: Comprehensive Analysis

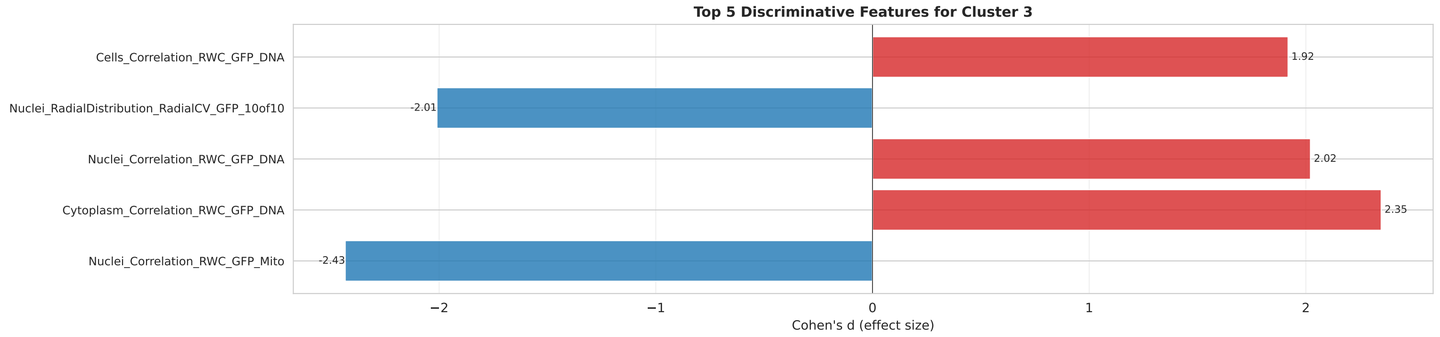
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CLUSTER 3 ANALYSIS
================================================================================================

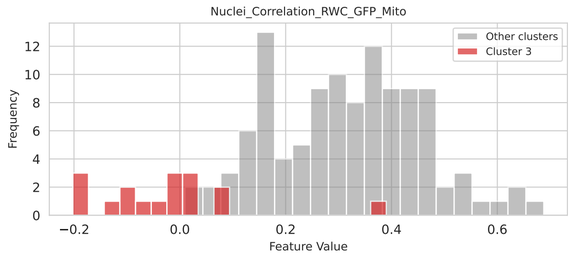
Size: 17 variants
Mean misloc_score: 0.964
Hit variants: 11 (64.7%)

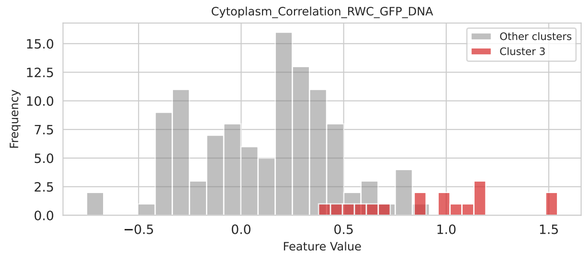
ALL VARIANTS (sorted by gene, position):
--------------------------------------------------
  BRCA1_Glu455Lys         |  DES_Arg355Pro          |  FXN_Met1Ile            |  HPRT1_Phe74Leu
  LITAF_Pro84Leu          |  LITAF_Trp116Gly        |  LITAF_Ser118Pro        |  LITAF_Leu122Val
  LITAF_Pro135Thr         |  LITAF_Pro135Arg        |  LITAF_Pro135Ser        |  LITAF_Phe136Leu
  LITAF_Val144Met         |  PLP1_Trp128Arg         |  PNPO_Arg95His          |  PNPO_Arg95Cys
  PNPO_Arg116Gln

DISEASE MODULES (detailed):
--------------------------------------------------
  Rare genetic neurological disorder: 16 [FXN_Met1Ile, LITAF_Pro135Thr, LITAF_Leu122Val, LITAF_Trp116Gly, PNPO_Arg95His, PLP1_Trp128Arg, PNPO_Arg95Cys, PNPO_Arg116Gln, DES_Arg355Pro, LITAF_Pro135Arg, LITAF_Phe136Leu, LITAF_Val144Met, LITAF_Pro135Ser, LITAF_Pro84Leu, LITAF_Ser118Pro, HPRT1_Phe74Leu]
  Rare inborn errors of metabolism: 4 [PNPO_Arg95His, PNPO_Arg95Cys, PNPO_Arg116Gln, HPRT1_Phe74Leu]
  Rare genetic eye disease: 2 [FXN_Met1Ile, PLP1_Trp128Arg]
  Rare genetic cardiac disease: 2 [FXN_Met1Ile, DES_Arg355Pro]
  Rare genetic renal disease: 2 [BRCA1_Glu455Lys, HPRT1_Phe74Leu]
  Rare genetic hematologic disease: 2 [BRCA1_Glu455Lys, HPRT1_Phe74Leu]
  Rare genetic developmental defect during embryogenesis: 1 [BRCA1_Glu455Lys]
  Rare genetic bone disease: 1 [BRCA1_Glu455Lys]
  Rare genetic gastroenterological disease: 1 [BRCA1_Glu455Lys]
  Rare genetic skin disease: 1 [BRCA1_Glu455Lys]

ANNOTATION ENRICHMENTS (values appearing ≥2 times):
--------------------------------------------------
  Disease: CHARCOT-MARIE-TOOTH DISEASE, DEMYELINATING, TYPE 1C (7x) [LITAF_Pro135Thr, LITAF_Leu122Val, LITAF_Trp116Gly, LITAF_Pro135Arg, LITAF_Val144Met, LITAF_Pro135Ser, LITAF_Pro84Leu]
  Disease: PYRIDOXAMINE 5-PRIME-PHOSPHATE OXIDASE DEFICIENCY (3x) [PNPO_Arg95His, PNPO_Arg95Cys, PNPO_Arg116Gln]
  inheritance_pattern: autosomal dominant (7x) [LITAF_Pro135Thr, LITAF_Leu122Val, LITAF_Trp116Gly, LITAF_Pro135Arg, LITAF_Val144Met, LITAF_Pro135Ser, LITAF_Pro84Leu]
  inheritance_pattern: autosomal recessive (4x) [FXN_Met1Ile, PNPO_Arg95His, PNPO_Arg95Cys, PNPO_Arg116Gln]
  inheritance_pattern: unknown (3x) [LITAF_Phe136Leu, LITAF_Ser118Pro, HPRT1_Phe74Leu]
  inheritance_pattern: multiple (2x) [BRCA1_Glu455Lys, DES_Arg355Pro]
  pLDDT_Category: 70-90 (8x) [LITAF_Pro135Thr, LITAF_Leu122Val, LITAF_Trp116Gly, LITAF_Pro135Arg, LITAF_Phe136Leu, LITAF_Pro135Ser, LITAF_Pro84Leu, LITAF_Ser118Pro]
  pLDDT_Category: >90 (6x) [PNPO_Arg95His, PLP1_Trp128Arg, PNPO_Arg95Cys, PNPO_Arg116Gln, DES_Arg355Pro, LITAF_Val144Met]
  pLDDT_Category: ⩽50 (2x) [FXN_Met1Ile, BRCA1_Glu455Lys]
  RSA_Category: 10-90 (10x) [LITAF_Leu122Val, LITAF_Trp116Gly, BRCA1_Glu455Lys, PNPO_Arg95His, PNPO_Arg95Cys, PNPO_Arg116Gln, DES_Arg355Pro, LITAF_Phe136Leu, LITAF_Pro84Leu, LITAF_Ser118Pro]
  RSA_Category: <10 (5x) [LITAF_Pro135Thr, PLP1_Trp128Arg, LITAF_Pro135Arg, LITAF_Val144Met, LITAF_Pro135Ser]
```







Cluster 4: Comprehensive Analysis

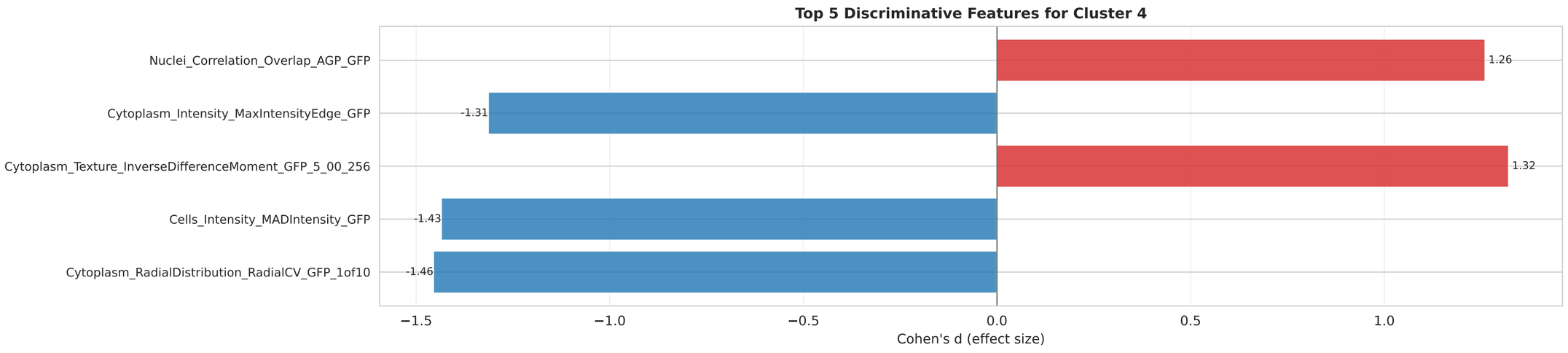
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CLUSTER 4 ANALYSIS
================================================================================================================

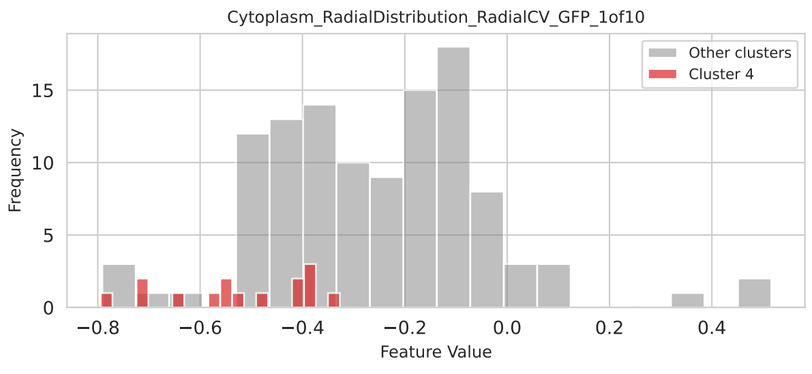
Size: 15 variants
Mean misloc_score: 0.968
Hit variants: 10 (66.7%)

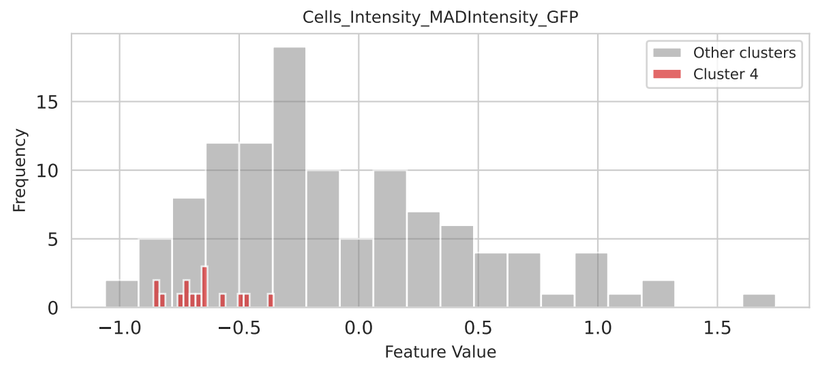
ALL VARIANTS (sorted by gene, position):
--------------------------------------------------
  DCX_Arg59His            |  DCX_Arg102Cys          |  DCX_Pro191Arg          |  F9_Cys28Arg
  FBP1_Gly164Ser          |  FBP1_Phe194Ser         |  FBP1_Gly260Arg         |  MVK_Leu255Pro
  NT5C3A_Cys108Arg        |  NT5C3A_Asp132Val       |  NT5C3A_Gly202Arg       |  NT5C3A_Gly275Arg
  PLP1_Gly74Arg           |  RHO_Pro267Thr          |  TTPA_Gly246Arg

DISEASE MODULES (detailed):
--------------------------------------------------
  Rare inborn errors of metabolism: 9 [NT5C3A_Gly275Arg, FBP1_Phe194Ser, MVK_Leu255Pro, FBP1_Gly164Ser, FBP1_Gly260Arg, NT5C3A_Asp132Val, NT5C3A_Gly202Arg, TTPA_Gly246Arg, NT5C3A_Cys108Arg]
  Rare genetic hematologic disease: 5 [NT5C3A_Gly275Arg, F9_Cys28Arg, NT5C3A_Asp132Val, NT5C3A_Gly202Arg, NT5C3A_Cys108Arg]
  Rare genetic neurological disorder: 5 [DCX_Arg102Cys, DCX_Pro191Arg, DCX_Arg59His, TTPA_Gly246Arg, PLP1_Gly74Arg]
  Rare genetic developmental defect during embryogenesis: 4 [DCX_Arg102Cys, MVK_Leu255Pro, DCX_Pro191Arg, DCX_Arg59His]
  Rare genetic eye disease: 4 [MVK_Leu255Pro, RHO_Pro267Thr, TTPA_Gly246Arg, PLP1_Gly74Arg]
  Rare genetic systemic or rheumatologic disease: 1 [MVK_Leu255Pro]
  Rare genetic immune disease: 1 [MVK_Leu255Pro]
  Rare genetic skin disease: 1 [MVK_Leu255Pro]
  Ciliopathy: 1 [RHO_Pro267Thr]

ANNOTATION ENRICHMENTS (values appearing ≥2 times):
--------------------------------------------------
  Disease: FRUCTOSE-1,6-BISPHOSPHATASE DEFICIENCY (3x) [FBP1_Phe194Ser, FBP1_Gly164Ser, FBP1_Gly260Arg]
  Disease: ANEMIA, CONGENITAL, NONSPHEROCYTIC HEMOLYTIC, 8 (2x) [NT5C3A_Gly275Arg, NT5C3A_Asp132Val]
  Disease: HEMOPHILIA B (2x) [F9_Cys28Arg, F9_Cys28Arg]
  inheritance_pattern: autosomal recessive (6x) [NT5C3A_Gly275Arg, FBP1_Phe194Ser, FBP1_Gly164Ser, FBP1_Gly260Arg, NT5C3A_Asp132Val, TTPA_Gly246Arg]
  inheritance_pattern: unknown (6x) [DCX_Arg102Cys, DCX_Pro191Arg, RHO_Pro267Thr, NT5C3A_Gly202Arg, NT5C3A_Cys108Arg, PLP1_Gly74Arg]
  pLDDT_Category: >90 (8x) [DCX_Arg102Cys, FBP1_Phe194Ser, MVK_Leu255Pro, RHO_Pro267Thr, DCX_Arg59His, FBP1_Gly164Ser, FBP1_Gly260Arg, TTPA_Gly246Arg]
  pLDDT_Category: 50-70 (2x) [DCX_Pro191Arg, F9_Cys28Arg]
  RSA_Category: 10-90 (6x) [DCX_Arg102Cys, MVK_Leu255Pro, RHO_Pro267Thr, F9_Cys28Arg, DCX_Arg59His, TTPA_Gly246Arg]
  RSA_Category: <10 (3x) [FBP1_Phe194Ser, FBP1_Gly164Ser, FBP1_Gly260Arg]
```







Cluster 5: Comprehensive Analysis

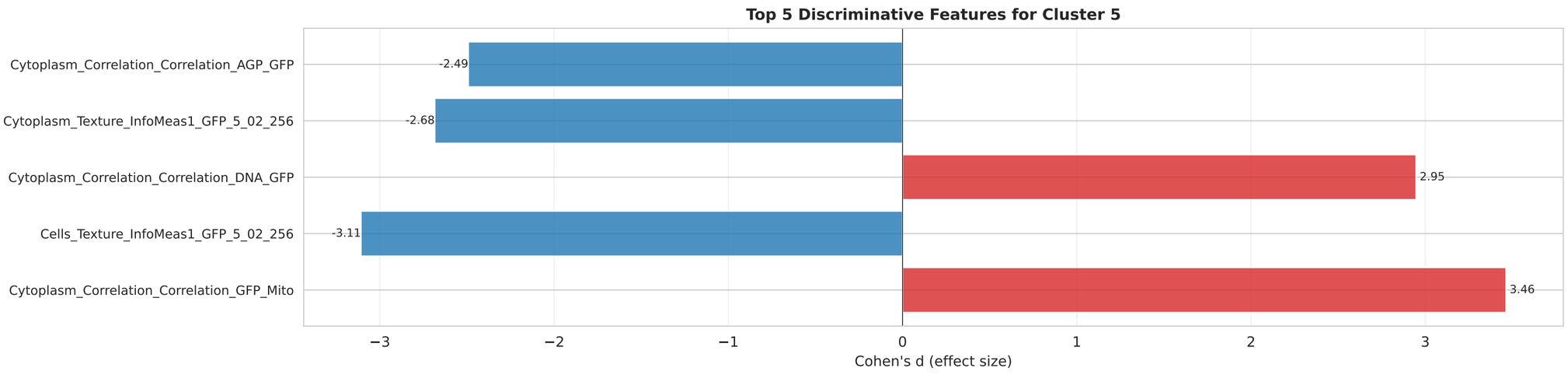
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CLUSTER 5 ANALYSIS
================================================================================================================

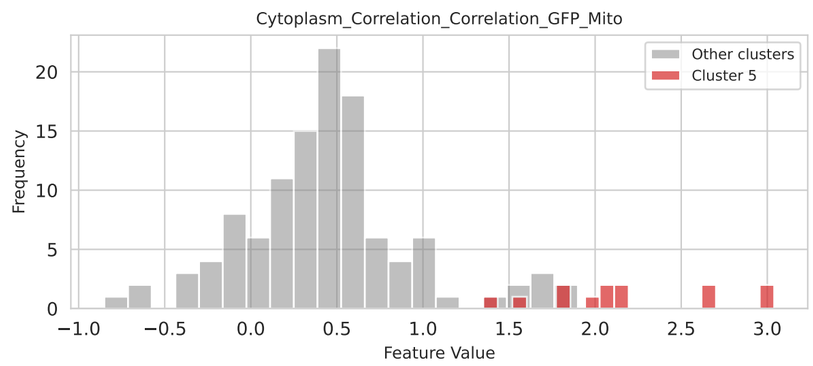
Size: 13 variants
Mean misloc_score: 0.983
Hit variants: 11 (84.6%)

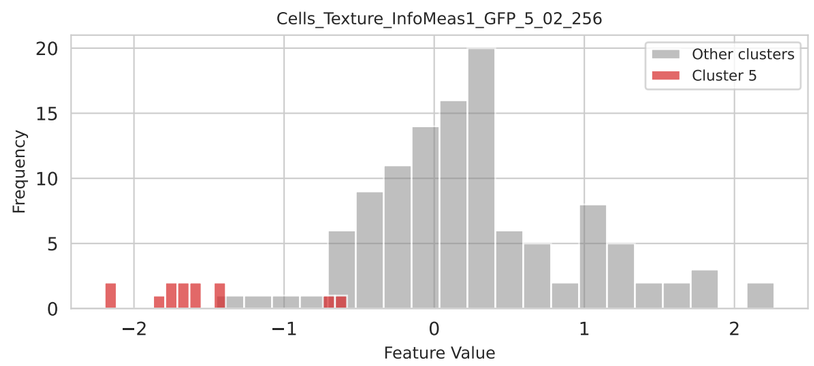
ALL VARIANTS (sorted by gene, position):
--------------------------------------------------
  PLP1_Thr43Ile           |  PLP1_Thr147Pro          |  PLP1_Ile152Thr          |  PLP1_Asp168His
  PLP1_Gly186Cys          |  PLP1_Leu189Pro          |  PLP1_Cys193Tyr          |  PMP22_Leu16Pro
  PMP22_Leu80Pro          |  PMP22_Cys109Arg         |  RHO_Cys110Tyr           |  RHO_Asp190Gly
  SFTPC_Leu188Gln

DISEASE MODULES (detailed):
--------------------------------------------------
  Rare genetic neurological disorder: 10 [PLP1_Gly186Cys, PLP1_Thr147Pro, PLP1_Asp168His, PMP22_Leu16Pro, PLP1_Cys193Tyr, PLP1_Ile152Thr, PLP1_Thr43Ile, PMP22_Leu80Pro, PMP22_Cys109Arg, PLP1_Leu189Pro]
  Rare genetic eye disease: 9 [PLP1_Gly186Cys, PLP1_Thr147Pro, PLP1_Asp168His, RHO_Cys110Tyr, PLP1_Cys193Tyr, PLP1_Ile152Thr, PLP1_Thr43Ile, RHO_Asp190Gly, PLP1_Leu189Pro]
  Genetic otorhinolaryngologic disease: 3 [PMP22_Leu16Pro, PMP22_Leu80Pro, PMP22_Cys109Arg]
  Rare chromosomal anomaly: 3 [PMP22_Leu16Pro, PMP22_Leu80Pro, PMP22_Cys109Arg]
  Ciliopathy: 2 [RHO_Cys110Tyr, RHO_Asp190Gly]
  Rare genetic respiratory disease: 1 [SFTPC_Leu188Gln]

ANNOTATION ENRICHMENTS (values appearing ≥2 times):
--------------------------------------------------
  Disease: PELIZAEUS-MERZBACHER DISEASE (5x) [PLP1_Thr147Pro, PLP1_Asp168His, PLP1_Ile152Thr, PLP1_Thr43Ile, PLP1_Leu189Pro]
  Disease: RETINITIS PIGMENTOSA 4 (2x) [RHO_Cys110Tyr, RHO_Asp190Gly]
  Disease: SPASTIC PARAPLEGIA 2, X-LINKED (2x) [PLP1_Ile152Thr, PLP1_Leu189Pro]
  inheritance_pattern: x-linked recessive (5x) [PLP1_Thr147Pro, PLP1_Asp168His, PLP1_Ile152Thr, PLP1_Thr43Ile, PLP1_Leu189Pro]
  inheritance_pattern: autosomal dominant (3x) [SFTPC_Leu188Gln, PMP22_Leu16Pro, PMP22_Leu80Pro]
  inheritance_pattern: unknown (3x) [PLP1_Gly186Cys, PLP1_Cys193Tyr, PMP22_Cys109Arg]
  inheritance_pattern: autosomal recessive (2x) [RHO_Cys110Tyr, RHO_Asp190Gly]
  pLDDT_Category: >90 (9x) [SFTPC_Leu188Gln, PLP1_Thr147Pro, PMP22_Leu16Pro, RHO_Cys110Tyr, PLP1_Cys193Tyr, PLP1_Thr43Ile, RHO_Asp190Gly, PMP22_Leu80Pro, PMP22_Cys109Arg]
  pLDDT_Category: 70-90 (4x) [PLP1_Gly186Cys, PLP1_Asp168His, PLP1_Ile152Thr, PLP1_Leu189Pro]
  RSA_Category: 10-90 (7x) [SFTPC_Leu188Gln, PLP1_Thr147Pro, PLP1_Asp168His, PMP22_Leu16Pro, PLP1_Thr43Ile, PMP22_Leu80Pro, PMP22_Cys109Arg]
  RSA_Category: <10 (6x) [PLP1_Gly186Cys, RHO_Cys110Tyr, PLP1_Cys193Tyr, PLP1_Ile152Thr, RHO_Asp190Gly, PLP1_Leu189Pro]
  Transmembrane (UniProt): Helical (3x) [PMP22_Leu16Pro, PMP22_Leu80Pro, PMP22_Cys109Arg]
  Topological domain (UniProt): Extracellular (3x) [RHO_Cys110Tyr, PLP1_Thr43Ile, RHO_Asp190Gly]
```







Cluster 6: Comprehensive Analysis

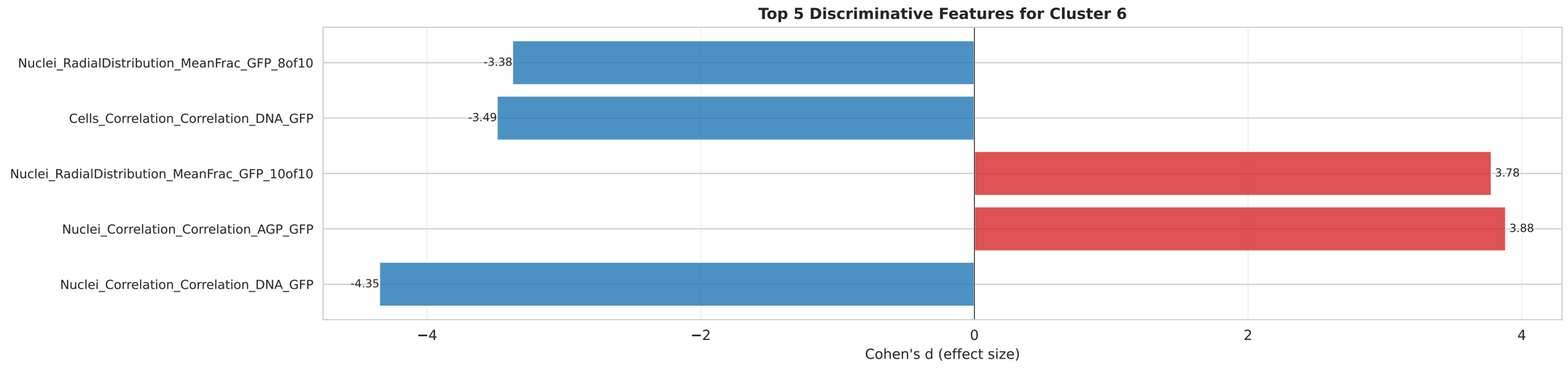
```
CLUSTER 6 ANALYSIS
================================================================================================================

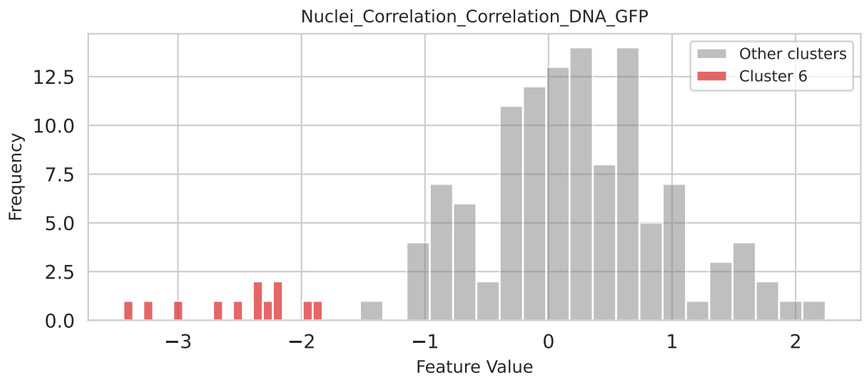
Size: 12 variants
Mean misloc_score: 0.992
Hit variants: 12 (100.0%)

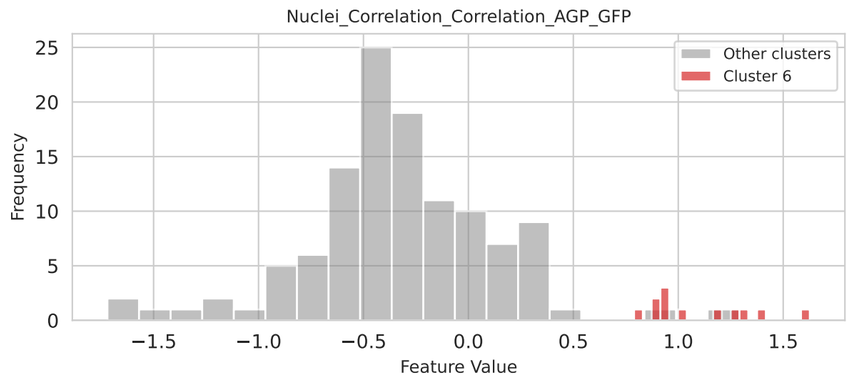
ALL VARIANTS (sorted by gene, position):
--------------------------------------------------
  AIPL1_Met79Thr          |  MLH1_Ile36Asn          |  MLH1_Cys39Tyr          |  MLH1_Ser44Phe
  MLH1_Gly147Arg          |  MLH1_Val185Gly         |  MLH1_Gln701His         |  NR0B1_Arg267Pro
  NR0B1_Arg425Ile         |  NR0B1_Asn440Ile        |  SMAD3_Trp94Leu         |  SMAD3_Cys109Tyr

DISEASE MODULES (detailed):
--------------------------------------------------
  Rare genetic immune disease: 6 [MLH1_Ser44Phe, MLH1_Gln701His, MLH1_Val185Gly, MLH1_Gly147Arg, MLH1_Cys39Tyr, MLH1_Ile36Asn]
  Rare genetic tumor: 6 [MLH1_Ser44Phe, MLH1_Gln701His, MLH1_Val185Gly, MLH1_Gly147Arg, MLH1_Cys39Tyr, MLH1_Ile36Asn]
  Inherited cancer-predisposing syndrome: 6 [MLH1_Ser44Phe, MLH1_Gln701His, MLH1_Val185Gly, MLH1_Gly147Arg, MLH1_Cys39Tyr, MLH1_Ile36Asn]
  Rare genetic gynecological and obstetrical diseases: 3 [NR0B1_Asn440Ile, NR0B1_Arg425Ile, NR0B1_Arg267Pro]
  Rare genetic cardiac disease: 3 [NR0B1_Asn440Ile, NR0B1_Arg425Ile, NR0B1_Arg267Pro]
  Rare genetic developmental defect during embryogenesis: 3 [NR0B1_Asn440Ile, NR0B1_Arg425Ile, NR0B1_Arg267Pro]
  Rare genetic urogenital disease: 3 [NR0B1_Asn440Ile, NR0B1_Arg425Ile, NR0B1_Arg267Pro]
  Rare genetic endocrine disease: 3 [NR0B1_Asn440Ile, NR0B1_Arg425Ile, NR0B1_Arg267Pro]
  Genetic infertility: 3 [NR0B1_Asn440Ile, NR0B1_Arg425Ile, NR0B1_Arg267Pro]
  Rare genetic skin disease: 3 [NR0B1_Asn440Ile, NR0B1_Arg425Ile, NR0B1_Arg267Pro]

ANNOTATION ENRICHMENTS (values appearing ≥2 times):
--------------------------------------------------
  Disease: LYNCH SYNDROME 2 (5x) [MLH1_Ser44Phe, MLH1_Gln701His, MLH1_Val185Gly, MLH1_Gly147Arg, MLH1_Cys39Tyr]
  Disease: ADRENAL HYPOPLASIA, CONGENITAL (4x) [NR0B1_Asn440Ile, NR0B1_Arg425Ile, NR0B1_Arg267Pro, NR0B1_Arg267Pro]
  Disease: LYNCH SYNDROME 1 (2x) [MLH1_Gln701His, MLH1_Cys39Tyr]
  inheritance_pattern: autosomal dominant (6x) [MLH1_Ser44Phe, MLH1_Gln701His, MLH1_Val185Gly, SMAD3_Trp94Leu, MLH1_Gly147Arg, MLH1_Cys39Tyr]
  inheritance_pattern: x-linked recessive (3x) [NR0B1_Asn440Ile, NR0B1_Arg425Ile, NR0B1_Arg267Pro]
  inheritance_pattern: unknown (3x) [SMAD3_Cys109Tyr, MLH1_Ile36Asn, AIPL1_Met79Thr]
  pLDDT_Category: >90 (10x) [MLH1_Ser44Phe, NR0B1_Asn440Ile, NR0B1_Arg425Ile, MLH1_Val185Gly, SMAD3_Trp94Leu, MLH1_Gly147Arg, MLH1_Cys39Tyr, SMAD3_Cys109Tyr, MLH1_Ile36Asn, AIPL1_Met79Thr]
  RSA_Category: <10 (8x) [MLH1_Ser44Phe, NR0B1_Arg425Ile, MLH1_Val185Gly, MLH1_Gly147Arg, MLH1_Cys39Tyr, SMAD3_Cys109Tyr, MLH1_Ile36Asn, AIPL1_Met79Thr]
  RSA_Category: 10-90 (3x) [NR0B1_Asn440Ile, NR0B1_Arg267Pro, SMAD3_Trp94Leu]
```







Cluster 7: Comprehensive Analysis

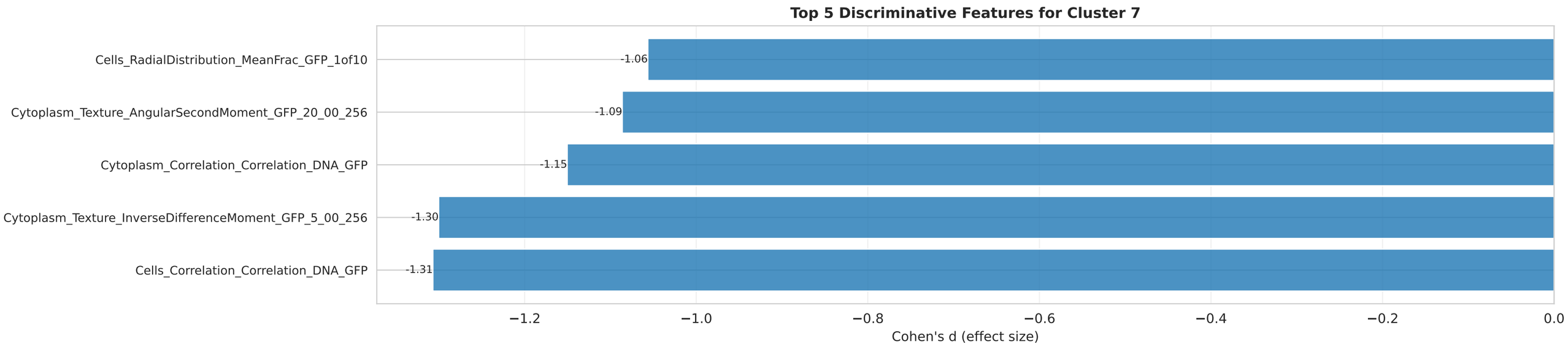
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CLUSTER 7 ANALYSIS
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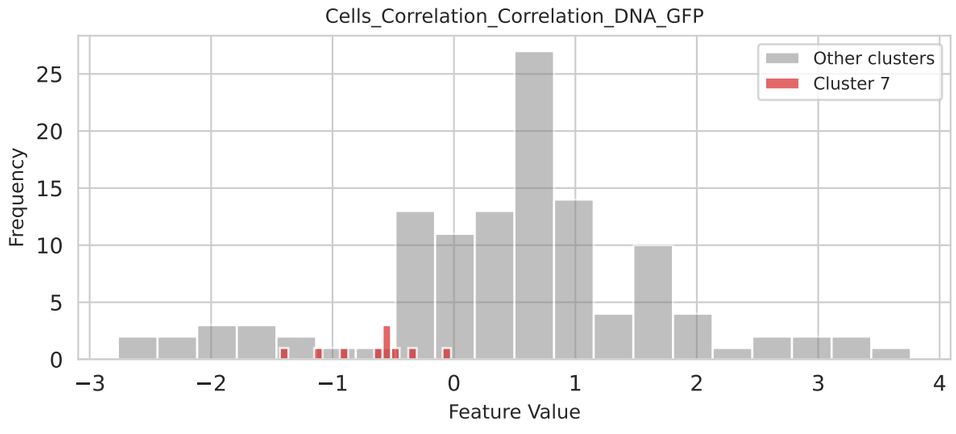
Size: 10 variants
Mean misloc_score: 0.954
Hit variants: 4 (40.0%)

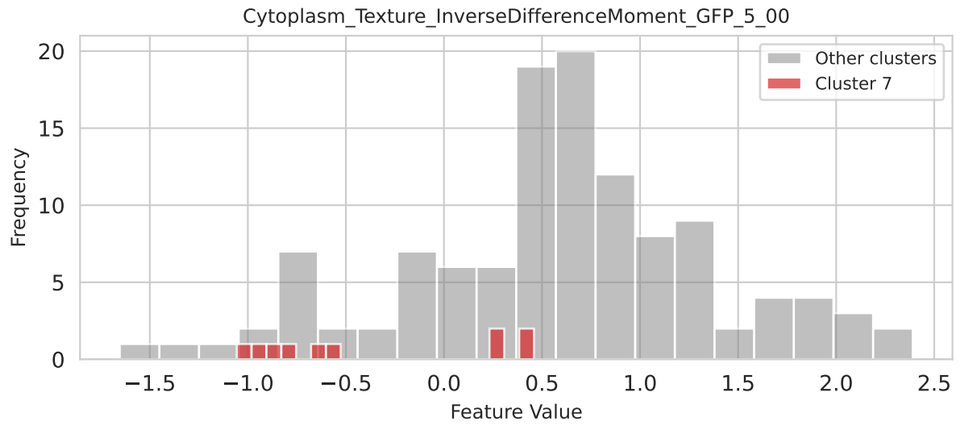
ALL VARIANTS (sorted by gene, position):
--------------------------------------------------
  BAG3_Leu462Pro          |  F9_Arg294Gln           |  F9_Ile316Thr           |  F9_Pro333Leu
  MLH1_Leu653Pro          |  NR0B1_Tyr380Asp        |  RET_Arg330Gln          |  SDHD_His102Leu
  SFTPA2_Phe198Ser        |  SFTPC_Ile73Thr

DISEASE MODULES (detailed):
------------------------------------------------
  Rare genetic developmental defect during embryogenesis: 3 [NR0B1_Tyr380Asp, SDHD_His102Leu, RET_Arg330Gln]
  Rare genetic endocrine disease: 3 [NR0B1_Tyr380Asp, SDHD_His102Leu, RET_Arg330Gln]
  Rare genetic hematologic disease: 3 [F9_Arg294Gln, F9_Pro333Leu, F9_Ile316Thr]
  Rare genetic neurological disorder: 3 [SDHD_His102Leu, BAG3_Leu462Pro, RET_Arg330Gln]
  Rare genetic tumor: 3 [SDHD_His102Leu, MLH1_Leu653Pro, RET_Arg330Gln]
  Inherited cancer-predisposing syndrome: 3 [SDHD_His102Leu, MLH1_Leu653Pro, RET_Arg330Gln]
  Rare genetic cardiac disease: 2 [NR0B1_Tyr380Asp, BAG3_Leu462Pro]
  Rare genetic renal disease: 2 [SDHD_His102Leu, RET_Arg330Gln]
  Rare genetic gynecological and obstetrical diseases: 1 [NR0B1_Tyr380Asp]
  Rare genetic urogenital disease: 1 [NR0B1_Tyr380Asp]

ANNOTATION ENRICHMENTS (values appearing ≥2 times):
------------------------------------------------
  Disease: HEMOPHILIA B (4x) [F9_Arg294Gln, F9_Arg294Gln, F9_Pro333Leu, F9_Ile316Thr]
  Disease: THROMBOPHILIA, X-LINKED, DUE TO FACTOR IX DEFECT (2x) [F9_Arg294Gln, F9_Ile316Thr]
  inheritance_pattern: autosomal dominant (5x) [SFTPA2_Phe198Ser, SFTPC_Ile73Thr, SDHD_His102Leu, BAG3_Leu462Pro, RET_Arg330Gln]
  inheritance_pattern: x-linked recessive (4x) [NR0B1_Tyr380Asp, F9_Arg294Gln, F9_Pro333Leu, F9_Ile316Thr]
  pLDDT_Category: >90 (8x) [SFTPA2_Phe198Ser, F9_Arg294Gln, F9_Pro333Leu, SDHD_His102Leu, MLH1_Leu653Pro, BAG3_Leu462Pro, F9_Ile316Thr, RET_Arg330Gln]
  RSA_Category: <10 (5x) [NR0B1_Tyr380Asp, SFTPA2_Phe198Ser, F9_Pro333Leu, MLH1_Leu653Pro, F9_Ile316Thr]
  RSA_Category: 10-90 (5x) [SFTPC_Ile73Thr, F9_Arg294Gln, SDHD_His102Leu, BAG3_Leu462Pro, RET_Arg330Gln]
```







Cluster 8: Comprehensive Analysis

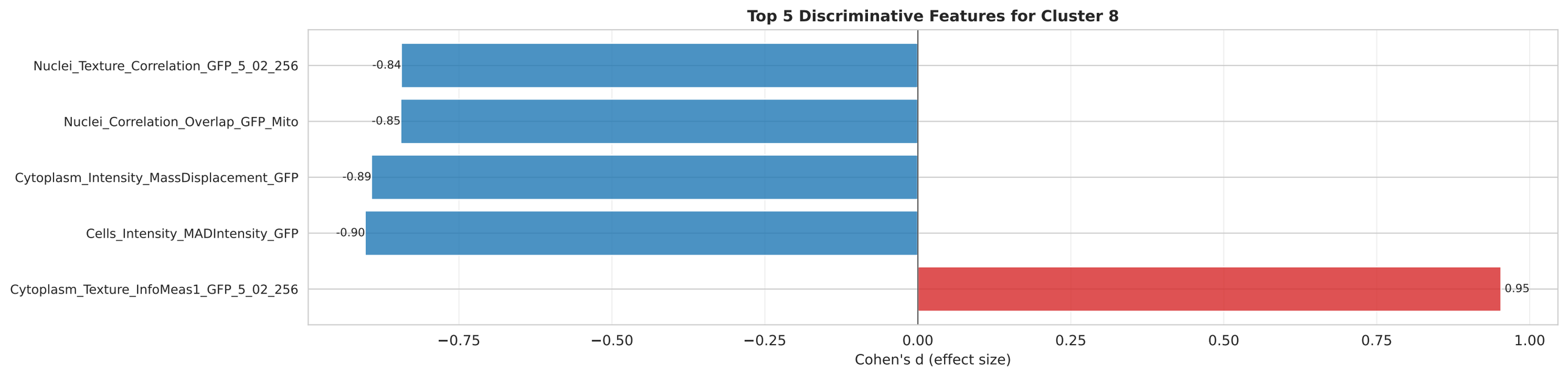
```
CLUSTER 8 ANALYSIS
================================================================================================================

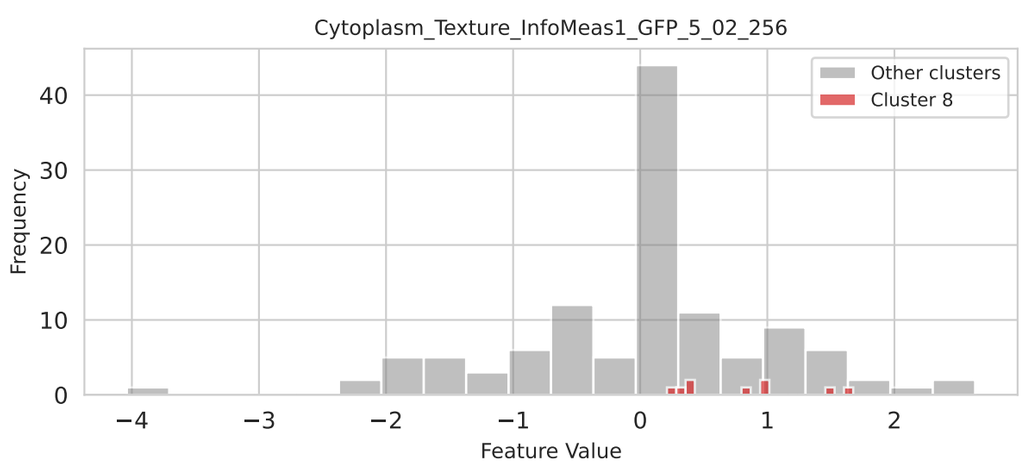
Size: 9 variants
Mean misloc_score: 0.951
Hit variants: 5 (55.6%)

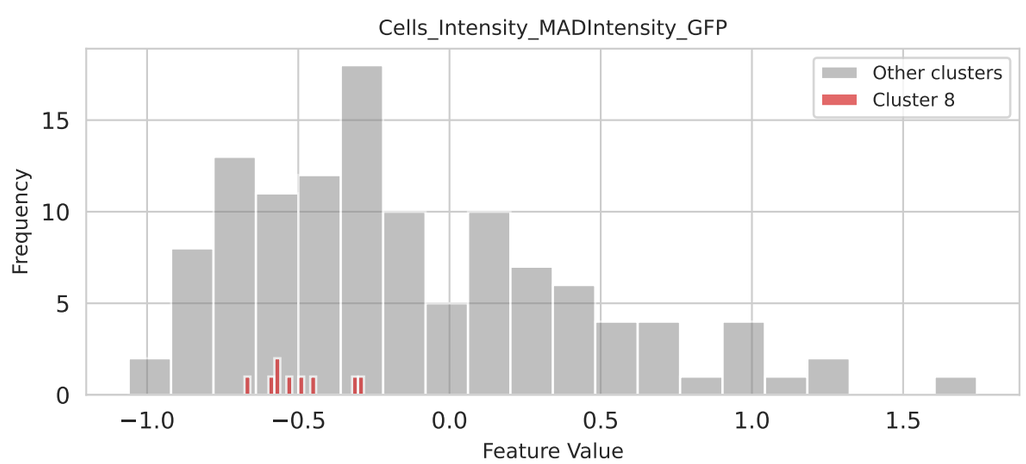
ALL VARIANTS (sorted by gene, position):
--------------------------------------------------
  AGXT_Gly116Arg          |  CDKN1A_Asp149Gly        |  FH_Asn340Lys            |  LITAF_Ala129Val
  SFTPA2_Gly231Val        |  SMAD3_Trp232Cys         |  SMAD3_Glu239Lys         |  SMAD3_Met255Arg
  SMAD3_Gly272Arg

DISEASE MODULES (detailed):
--------------------------------------------------
  Rare genetic vascular disease: 4 [SMAD3_Glu239Lys, SMAD3_Gly272Arg, SMAD3_Met255Arg, SMAD3_Trp232Cys]
  Rare genetic systemic or rheumatologic disease: 4 [SMAD3_Glu239Lys, SMAD3_Gly272Arg, SMAD3_Met255Arg, SMAD3_Trp232Cys]
  Rare genetic renal disease: 2 [AGXT_Gly116Arg, FH_Asn340Lys]
  Rare inborn errors of metabolism: 2 [AGXT_Gly116Arg, FH_Asn340Lys]
  Rare genetic endocrine disease: 2 [FH_Asn340Lys, CDKN1A_Asp149Gly]
  Rare genetic neurological disorder: 2 [FH_Asn340Lys, LITAF_Ala129Val]
  Rare genetic tumor: 2 [FH_Asn340Lys, CDKN1A_Asp149Gly]
  Inherited cancer-predisposing syndrome: 2 [FH_Asn340Lys, CDKN1A_Asp149Gly]
  Rare genetic eye disease: 1 [AGXT_Gly116Arg]

ANNOTATION ENRICHMENTS (values appearing ≥2 times):
--------------------------------------------------
  Disease: LOEYS-DIETZ SYNDROME 3 (2x) [SMAD3_Glu239Lys, SMAD3_Gly272Arg]
  inheritance_pattern: autosomal dominant (4x) [SMAD3_Glu239Lys, SMAD3_Gly272Arg, LITAF_Ala129Val, SFTPA2_Gly231Val]
  inheritance_pattern: unknown (3x) [SMAD3_Met255Arg, SMAD3_Trp232Cys, CDKN1A_Asp149Gly]
  pLDDT_Category: >90 (7x) [AGXT_Gly116Arg, SMAD3_Glu239Lys, FH_Asn340Lys, SMAD3_Gly272Arg, SMAD3_Met255Arg, SMAD3_Trp232Cys, SFTPA2_Gly231Val]
  RSA_Category: <10 (5x) [AGXT_Gly116Arg, SMAD3_Glu239Lys, SMAD3_Met255Arg, SMAD3_Trp232Cys, SFTPA2_Gly231Val]
  RSA_Category: 10-90 (2x) [FH_Asn340Lys, SMAD3_Gly272Arg]
```







Cluster 9: Comprehensive Analysis

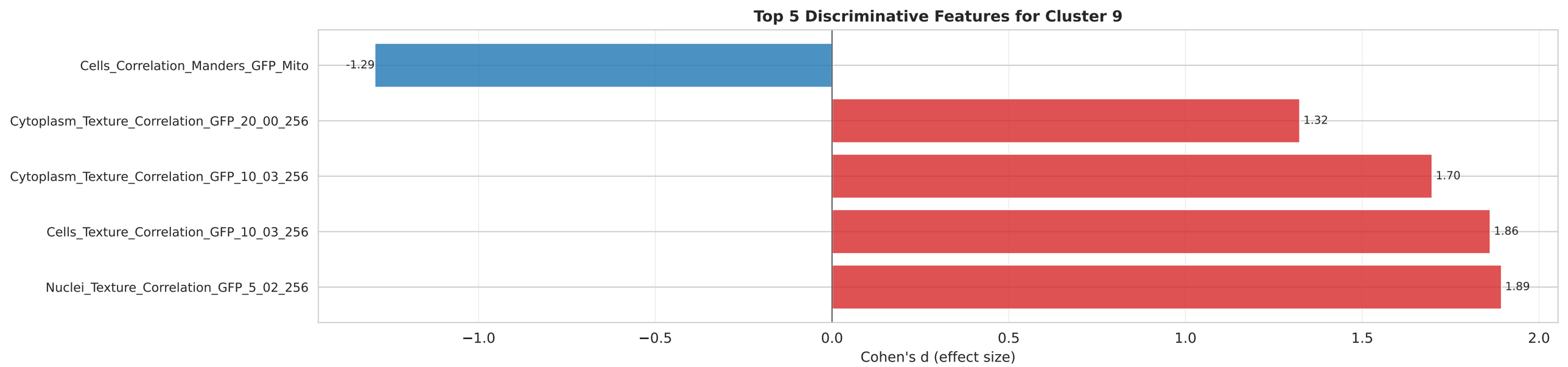
```
CLUSTER 9 ANALYSIS
================================================================================================================

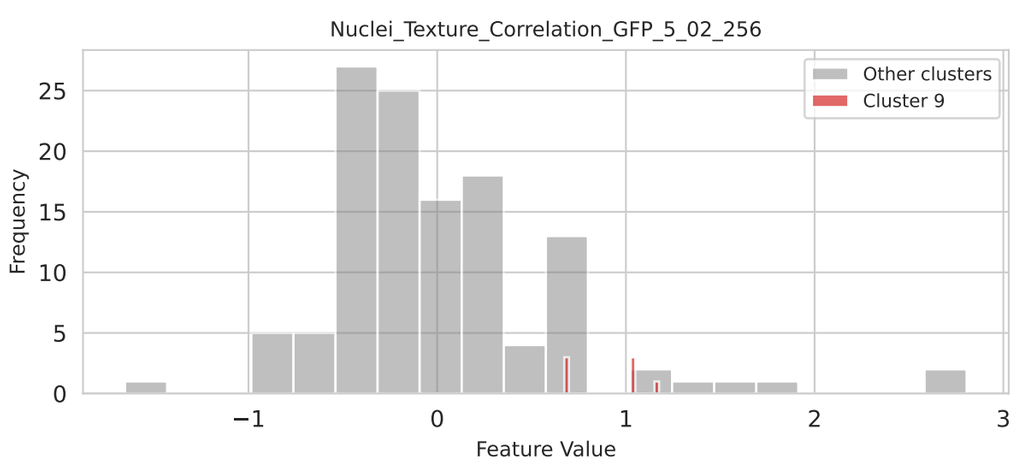
Size: 7 variants
Mean misloc_score: 0.933
Hit variants: 4 (57.1%)

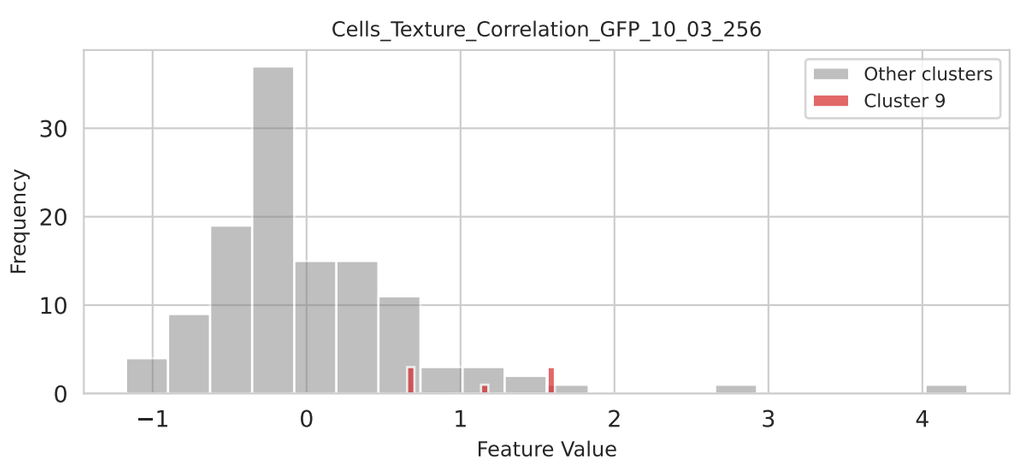
ALL VARIANTS (sorted by gene, position):
--------------------------------------------------
  BRCA1_Met1Ile           |  BRCA1_Met1Thr           |  BRCA1_Ala102Gly         |  FOXP3_Leu242Pro
  MLH1_Ser295Thr          |  MLH1_Asp485Glu          |  MLH1_Lys751Arg

DISEASE MODULES (detailed):
--------------------------------------------------
  Rare genetic tumor: 6 [BRCA1_Met1Ile, BRCA1_Met1Thr, BRCA1_Ala102Gly, MLH1_Lys751Arg, MLH1_Ser295Thr, MLH1_Asp485Glu]
  Inherited cancer-predisposing syndrome: 6 [BRCA1_Met1Ile, BRCA1_Met1Thr, BRCA1_Ala102Gly, MLH1_Lys751Arg, MLH1_Ser295Thr, MLH1_Asp485Glu]
  Rare genetic gastroenterological disease: 4 [BRCA1_Met1Ile, BRCA1_Met1Thr, BRCA1_Ala102Gly, FOXP3_Leu242Pro]
  Rare genetic immune disease: 4 [MLH1_Lys751Arg, MLH1_Ser295Thr, FOXP3_Leu242Pro, MLH1_Asp485Glu]
  Rare genetic renal disease: 3 [BRCA1_Met1Ile, BRCA1_Met1Thr, BRCA1_Ala102Gly]
  Rare genetic developmental defect during embryogenesis: 3 [BRCA1_Met1Ile, BRCA1_Met1Thr, BRCA1_Ala102Gly]
  Rare genetic bone disease: 3 [BRCA1_Met1Ile, BRCA1_Met1Thr, BRCA1_Ala102Gly]
  Rare genetic hematologic disease: 3 [BRCA1_Met1Ile, BRCA1_Met1Thr, BRCA1_Ala102Gly]
  Rare genetic skin disease: 3 [BRCA1_Met1Ile, BRCA1_Met1Thr, BRCA1_Ala102Gly]
  Rare genetic endocrine disease: 1 [FOXP3_Leu242Pro]

ANNOTATION ENRICHMENTS (values appearing ≥2 times):
--------------------------------------------------
  Disease: BREAST-OVARIAN CANCER, FAMILIAL, SUSCEPTIBILITY TO, 1 (4x) [BRCA1_Met1Ile, BRCA1_Met1Ile, BRCA1_Met1Thr, BRCA1_Ala102Gly]
  Disease: LYNCH SYNDROME 2 (3x) [MLH1_Lys751Arg, MLH1_Lys751Arg, MLH1_Ser295Thr]
  inheritance_pattern: multiple (4x) [BRCA1_Met1Ile, BRCA1_Met1Thr, BRCA1_Ala102Gly, MLH1_Lys751Arg]
  inheritance_pattern: unknown (2x) [FOXP3_Leu242Pro, MLH1_Asp485Glu]
  pLDDT_Category: ⩽50 (2x) [BRCA1_Met1Ile, BRCA1_Met1Thr]
  RSA_Category: ⩾90 (2x) [BRCA1_Met1Ile, BRCA1_Met1Thr]
  RSA_Category: 10-90 (2x) [BRCA1_Ala102Gly, MLH1_Lys751Arg]
```







Cluster 10: Comprehensive Analysis

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CLUSTER 10 ANALYSIS
================================================================================================================

Size: 7 variants
Mean misloc_score: 0.982
Hit variants: 1 (14.3%)

ALL VARIANTS (sorted by gene, position):
--------------------------------------------------
  DCX_Ser129Leu           |  MVK_Leu41Pro           |  MVK_Ser329Asn          |  RP2_Cys86Tyr
  RP2_Leu253Arg           |  TAT_Arg119Trp          |  TAT_Leu273Pro

DISEASE MODULES (detailed):
--------------------------------------------------
  Rare genetic eye disease: 6 [MVK_Leu41Pro, TAT_Arg119Trp, RP2_Cys86Tyr, MVK_Ser329Asn, RP2_Leu253Arg, TAT_Leu273Pro]
  Rare inborn errors of metabolism: 4 [MVK_Leu41Pro, TAT_Arg119Trp, MVK_Ser329Asn, TAT_Leu273Pro]
  Rare genetic skin disease: 4 [MVK_Leu41Pro, TAT_Arg119Trp, MVK_Ser329Asn, TAT_Leu273Pro]
  Rare genetic developmental defect during embryogenesis: 3 [DCX_Ser129Leu, MVK_Leu41Pro, MVK_Ser329Asn]
  Rare genetic systemic or rheumatologic disease: 2 [MVK_Leu41Pro, MVK_Ser329Asn]
  Rare genetic immune disease: 2 [MVK_Leu41Pro, MVK_Ser329Asn]
  Ciliopathy: 2 [RP2_Cys86Tyr, RP2_Leu253Arg]
  Rare genetic neurological disorder: 1 [DCX_Ser129Leu]

ANNOTATION ENRICHMENTS (values appearing ≥2 times):
--------------------------------------------------
  Disease: POROKERATOSIS 3, MULTIPLE TYPES (2x) [MVK_Leu41Pro, MVK_Ser329Asn]
  Disease: HYPER-IgD SYNDROME (2x) [MVK_Ser329Asn, MVK_Ser329Asn]
  inheritance_pattern: unknown (4x) [DCX_Ser129Leu, RP2_Cys86Tyr, RP2_Leu253Arg, TAT_Leu273Pro]
  pLDDT_Category: >90 (5x) [MVK_Leu41Pro, TAT_Arg119Trp, RP2_Cys86Tyr, MVK_Ser329Asn, RP2_Leu253Arg]
  RSA_Category: <10 (5x) [DCX_Ser129Leu, MVK_Leu41Pro, RP2_Cys86Tyr, MVK_Ser329Asn, RP2_Leu253Arg]
```

